Correctness Validation

The first step in the evaluation is to validate the correctness of the implementation. This involves providing structured evidence that the user-space framework faithfully implements OLSR-style semantics and that the resulting data plane behavior aligns with the computed routes.

All correctness validation checks were performed on the fixed, five-node logical topology that was constructed using software-based isolation rules, as detailed in Section 3.2.3. To provide a clear frame of reference for the results that follow, this topology is illustrated again in Figure 4.1. The key characteristics of this topology are the specific one-hop adjacencies:

- Node 1 has only one direct neighbor: Node 2.
- Node 2 has three direct neighbors: Nodes 1, 3, and 4.
- Node 3 has three direct neighbors: Nodes 2, 4, and 5.
- Node 4 has two direct neighbors: Nodes 2 and 3.
- Node 5 has only one direct neighbor: Node 3.

This specific structure is essential for verifying the protocol's behavior across single-hop, two-hop, and three-hop paths.

The primary source of evidence for this validation is the real-time `PrettyTable` dashboard generated by each node. Figures 4.2(a) through 4.2(e) present the dashboard outputs captured from all five nodes after the network had stabilized on this topology. These five screenshots serve as the ground truth for the analysis in this section and will be referred to frequently to verify the protocol's behavior against the expected state defined by Figure 4.1.

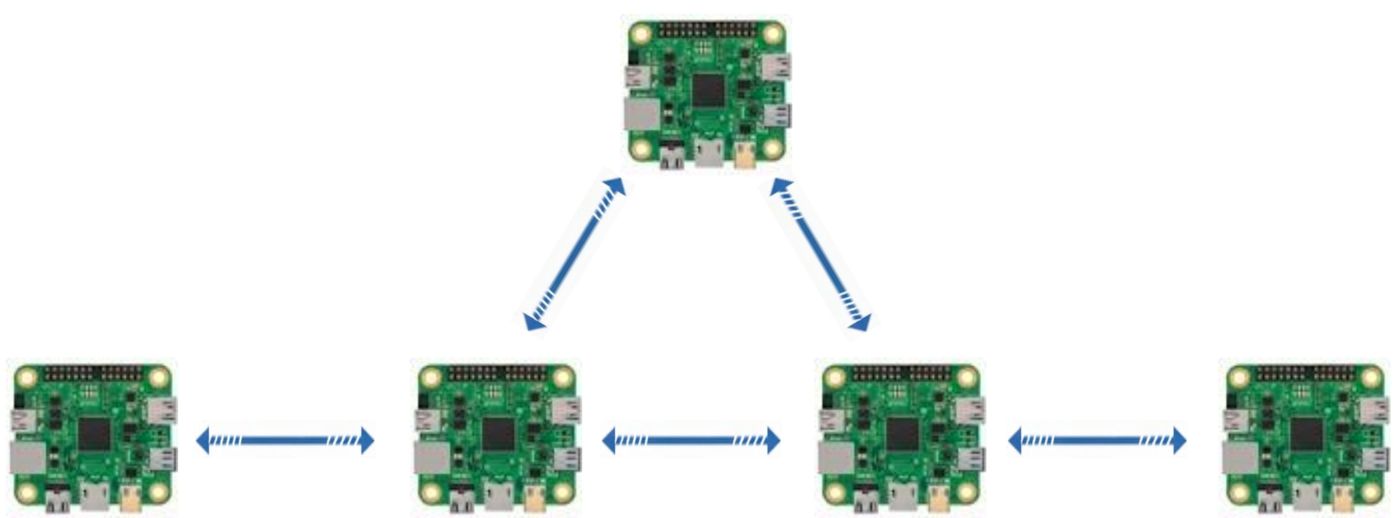
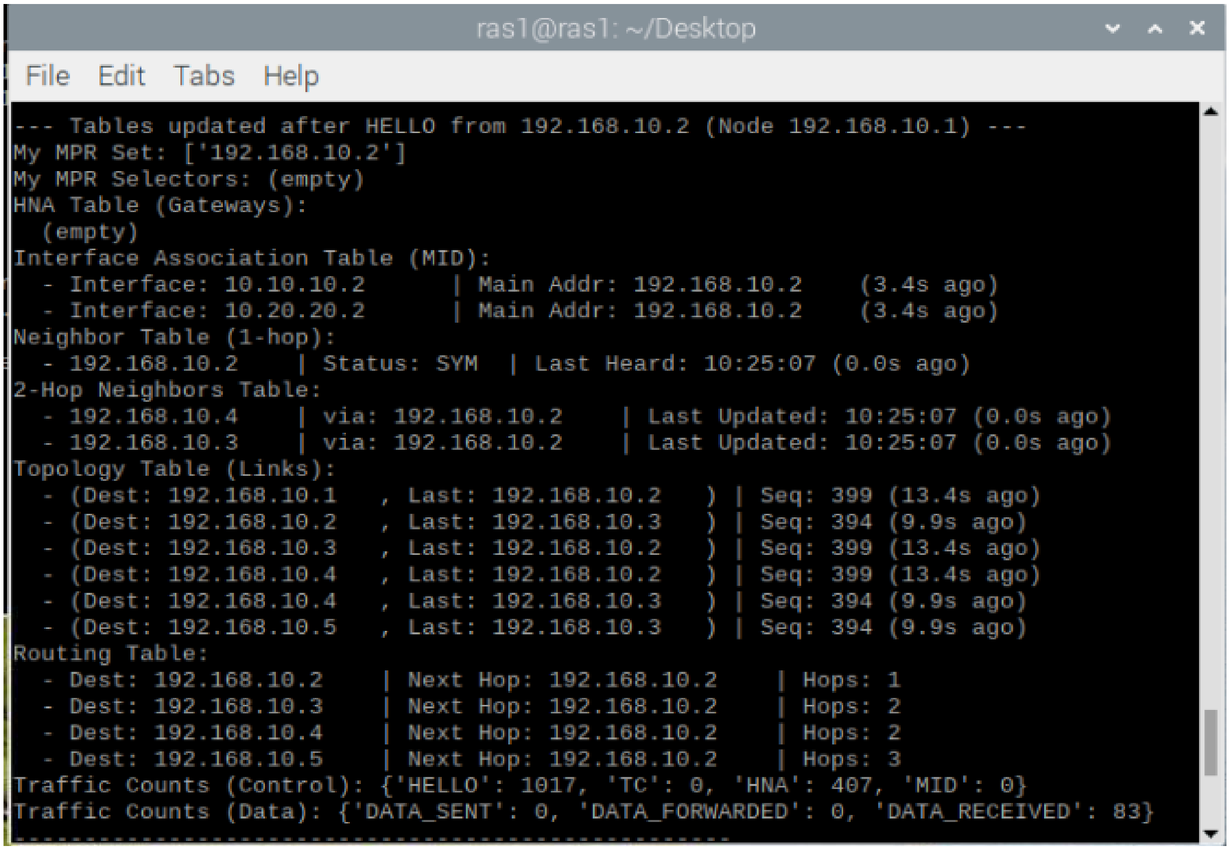
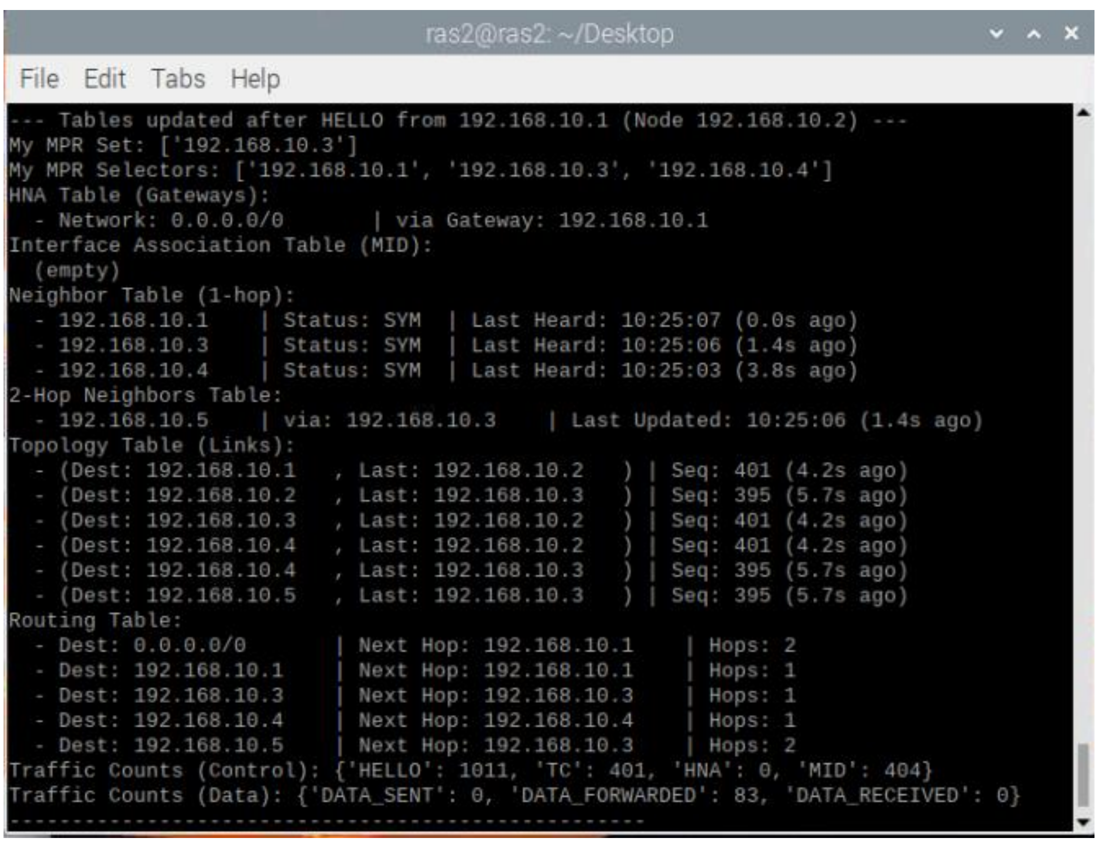


Figure 1:The Baseline Experimental Topology



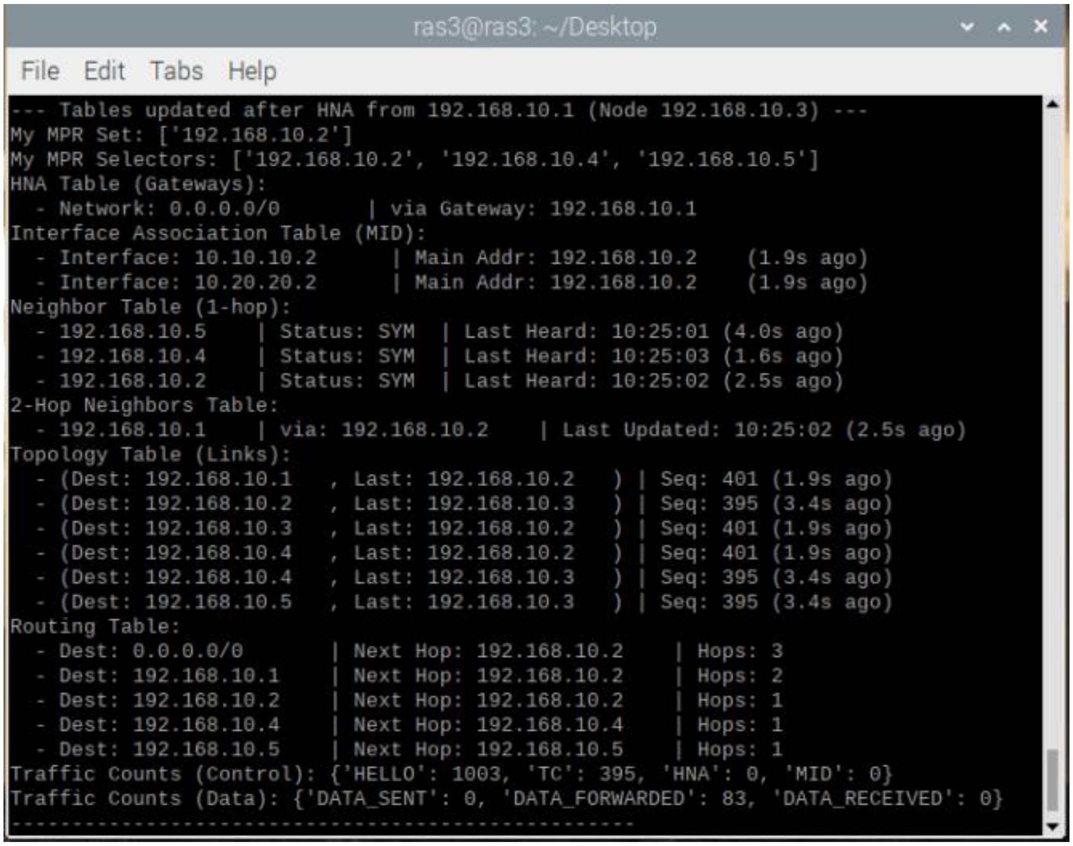
```
ras1@ras1: ~/Desktop
File  Edit  Tabs  Help
--- Tables updated after HELLO from 192.168.10.2 (Node 192.168.10.1) ---
My MPR Set: ['192.168.10.2']
My MPR Selectors: (empty)
HNA Table (Gateways):
  (empty)
Interface Association Table (MID):
  - Interface: 10.10.10.2      | Main Addr: 192.168.10.2    (3.4s ago)
  - Interface: 10.20.20.2      | Main Addr: 192.168.10.2    (3.4s ago)
Neighbor Table (1-hop):
  - 192.168.10.2    | Status: SYM  | Last Heard: 10:25:07 (0.0s ago)
2-Hop Neighbors Table:
  - 192.168.10.4    | via: 192.168.10.2    | Last Updated: 10:25:07 (0.0s ago)
  - 192.168.10.3    | via: 192.168.10.2    | Last Updated: 10:25:07 (0.0s ago)
Topology Table (Links):
  - (Dest: 192.168.10.1   , Last: 192.168.10.2   ) | Seq: 399 (13.4s ago)
  - (Dest: 192.168.10.2   , Last: 192.168.10.3   ) | Seq: 394 (9.9s ago)
  - (Dest: 192.168.10.3   , Last: 192.168.10.2   ) | Seq: 399 (13.4s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.2   ) | Seq: 399 (13.4s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.3   ) | Seq: 394 (9.9s ago)
  - (Dest: 192.168.10.5   , Last: 192.168.10.3   ) | Seq: 394 (9.9s ago)
Routing Table:
  - Dest: 192.168.10.2     | Next Hop: 192.168.10.2     | Hops: 1
  - Dest: 192.168.10.3     | Next Hop: 192.168.10.2     | Hops: 2
  - Dest: 192.168.10.4     | Next Hop: 192.168.10.2     | Hops: 2
  - Dest: 192.168.10.5     | Next Hop: 192.168.10.2     | Hops: 3
Traffic Counts (Control): {'HELLO': 1017, 'TC': 0, 'HNA': 407, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 83}
```

Figure (a): Node State Dashboard for Node 1(192.168.10.1)



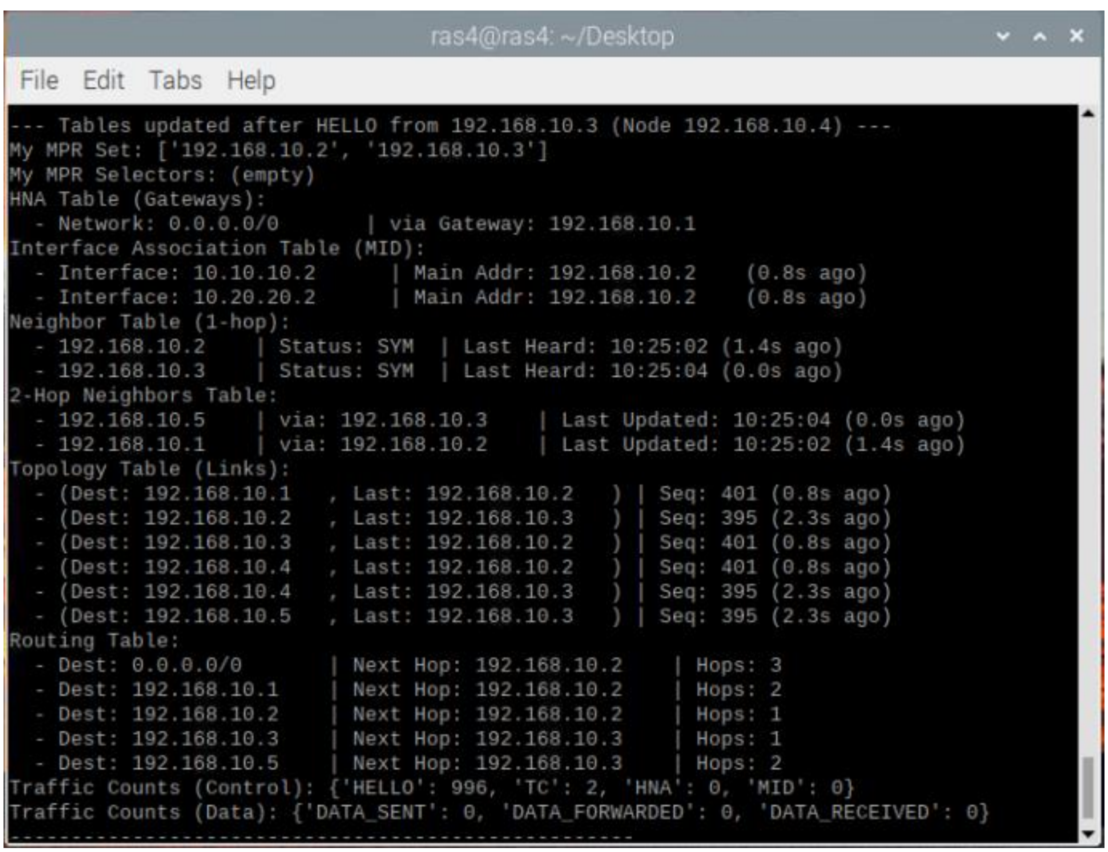
```
ras2@ras2: ~/Desktop
File  Edit  Tabs  Help
--- Tables updated after HELLO from 192.168.10.1 (Node 192.168.10.2) ---
My MPR Set: ['192.168.10.3']
My MPR Selectors: ['192.168.10.1', '192.168.10.3', '192.168.10.4']
HNA Table (Gateways):
  - Network: 0.0.0.0/0       | via Gateway: 192.168.10.1
Interface Association Table (MID):
  (empty)
Neighbor Table (1-hop):
  - 192.168.10.1    | Status: SYM  | Last Heard: 10:25:07 (0.0s ago)
  - 192.168.10.3    | Status: SYM  | Last Heard: 10:25:06 (1.4s ago)
  - 192.168.10.4    | Status: SYM  | Last Heard: 10:25:03 (3.8s ago)
2-Hop Neighbors Table:
  - 192.168.10.5    | via: 192.168.10.3    | Last Updated: 10:25:06 (1.4s ago)
Topology Table (Links):
  - (Dest: 192.168.10.1   , Last: 192.168.10.2   ) | Seq: 401 (4.2s ago)
  - (Dest: 192.168.10.2   , Last: 192.168.10.3   ) | Seq: 395 (5.7s ago)
  - (Dest: 192.168.10.3   , Last: 192.168.10.2   ) | Seq: 401 (4.2s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.2   ) | Seq: 401 (4.2s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.3   ) | Seq: 395 (5.7s ago)
  - (Dest: 192.168.10.5   , Last: 192.168.10.3   ) | Seq: 395 (5.7s ago)
Routing Table:
  - Dest: 0.0.0.0/0        | Next Hop: 192.168.10.1     | Hops: 2
  - Dest: 192.168.10.1     | Next Hop: 192.168.10.1     | Hops: 1
  - Dest: 192.168.10.3     | Next Hop: 192.168.10.3     | Hops: 1
  - Dest: 192.168.10.4     | Next Hop: 192.168.10.4     | Hops: 1
  - Dest: 192.168.10.5     | Next Hop: 192.168.10.3     | Hops: 2
Traffic Counts (Control): {'HELLO': 1011, 'TC': 401, 'HNA': 0, 'MID': 404}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 83, 'DATA_RECEIVED': 0}
```

Figure (b): Node State Dashboard for Node 2(192.168.10.2)



```
ras3@ras3: ~/Desktop
File  Edit  Tabs  Help
--- Tables updated after HNA from 192.168.10.1 (Node 192.168.10.3) ---
My MPR Set: ['192.168.10.2']
My MPR Selectors: ['192.168.10.2', '192.168.10.4', '192.168.10.5']
HNA Table (Gateways):
  - Network: 0.0.0.0/0       | via Gateway: 192.168.10.1
Interface Association Table (MID):
  - Interface: 10.10.10.2      | Main Addr: 192.168.10.2    (1.9s ago)
  - Interface: 10.20.20.2      | Main Addr: 192.168.10.2    (1.9s ago)
Neighbor Table (1-hop):
  - 192.168.10.5    | Status: SYM  | Last Heard: 10:25:01 (4.0s ago)
  - 192.168.10.4    | Status: SYM  | Last Heard: 10:25:03 (1.6s ago)
  - 192.168.10.2    | Status: SYM  | Last Heard: 10:25:02 (2.5s ago)
2-Hop Neighbors Table:
  - 192.168.10.1    | via: 192.168.10.2    | Last Updated: 10:25:02 (2.5s ago)
Topology Table (Links):
  - (Dest: 192.168.10.1   , Last: 192.168.10.2   ) | Seq: 401 (1.9s ago)
  - (Dest: 192.168.10.2   , Last: 192.168.10.3   ) | Seq: 395 (3.4s ago)
  - (Dest: 192.168.10.3   , Last: 192.168.10.2   ) | Seq: 401 (1.9s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.2   ) | Seq: 401 (1.9s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.3   ) | Seq: 395 (3.4s ago)
  - (Dest: 192.168.10.5   , Last: 192.168.10.3   ) | Seq: 395 (3.4s ago)
Routing Table:
  - Dest: 0.0.0.0/0       | Next Hop: 192.168.10.2    | Hops: 3
  - Dest: 192.168.10.1    | Next Hop: 192.168.10.2    | Hops: 2
  - Dest: 192.168.10.2    | Next Hop: 192.168.10.2    | Hops: 1
  - Dest: 192.168.10.4    | Next Hop: 192.168.10.4    | Hops: 1
  - Dest: 192.168.10.5    | Next Hop: 192.168.10.5    | Hops: 1
Traffic Counts (Control): {'HELLO': 1003, 'TC': 395, 'HNA': 0, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 83, 'DATA_RECEIVED': 0}
---------------------------------------------------
```

Figure (c): Node State Dashboard for Node 3(192.168.10.3)



```
ras4@ras4: ~/Desktop
File  Edit  Tabs  Help
--- Tables updated after HELLO from 192.168.10.3 (Node 192.168.10.4) ---
My MPR Set: ['192.168.10.2', '192.168.10.3']
My MPR Selectors: (empty)
HNA Table (Gateways):
  - Network: 0.0.0.0/0       | via Gateway: 192.168.10.1
Interface Association Table (MID):
  - Interface: 10.10.10.2      | Main Addr: 192.168.10.2    (0.8s ago)
  - Interface: 10.20.20.2      | Main Addr: 192.168.10.2    (0.8s ago)
Neighbor Table (1-hop):
  - 192.168.10.2    | Status: SYM  | Last Heard: 10:25:02 (1.4s ago)
  - 192.168.10.3    | Status: SYM  | Last Heard: 10:25:04 (0.0s ago)
2-Hop Neighbors Table:
  - 192.168.10.5    | via: 192.168.10.3    | Last Updated: 10:25:04 (0.0s ago)
  - 192.168.10.1    | via: 192.168.10.2    | Last Updated: 10:25:02 (1.4s ago)
Topology Table (Links):
  - (Dest: 192.168.10.1   , Last: 192.168.10.2   ) | Seq: 401 (0.8s ago)
  - (Dest: 192.168.10.2   , Last: 192.168.10.3   ) | Seq: 395 (2.3s ago)
  - (Dest: 192.168.10.3   , Last: 192.168.10.2   ) | Seq: 401 (0.8s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.2   ) | Seq: 401 (0.8s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.3   ) | Seq: 395 (2.3s ago)
  - (Dest: 192.168.10.5   , Last: 192.168.10.3   ) | Seq: 395 (2.3s ago)
Routing Table:
  - Dest: 0.0.0.0/0       | Next Hop: 192.168.10.2    | Hops: 3
  - Dest: 192.168.10.1    | Next Hop: 192.168.10.2    | Hops: 2
  - Dest: 192.168.10.2    | Next Hop: 192.168.10.2    | Hops: 1
  - Dest: 192.168.10.3    | Next Hop: 192.168.10.3    | Hops: 1
  - Dest: 192.168.10.5    | Next Hop: 192.168.10.3    | Hops: 2
Traffic Counts (Control): {'HELLO': 996, 'TC': 2, 'HNA': 0, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
---------------------------------------------------
```

Figure (d): Node State Dashboard for Node 4(192.168.10.4)

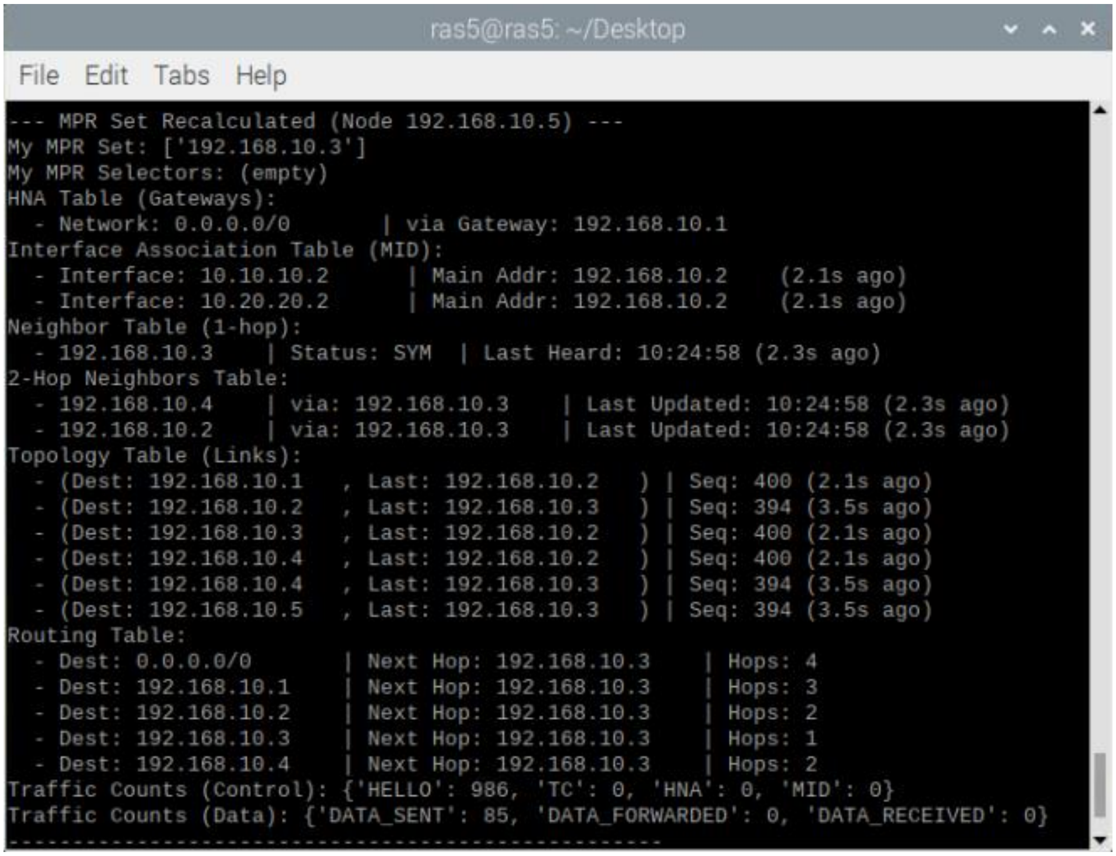


Figure (e): Node State Dashboard for Node 5(192.168.10.5)

1.1.1 Neighbor Table Accuracy and Two-Hop Coverage

Purpose: To verify that the HELLO message processing logic correctly discovers and maintains one-hop neighbors, and by extension, correctly identifies all reachable two-hop neighbors.

Validation Method: The `Neighbor Table (1-hop)` and 2-Hop Neighbors Table sections of the live dashboard on each node were inspected and compared against the known logical topology (Figure 4.1).

Observations: The dashboards confirmed that, after the warm-up period (2 – 3 Hello interval), the neighbor tables on all nodes perfectly matched the logical connectivity defined by the isolation rules. Specifically:

- Node 1 correctly listed only Node 2, in it's `Neighbor Table (1-hop)`, as its symmetric neighbor see Figure 4.2 (a).
- Node 2 listed Nodes 1, 3, and 4 see Figure 4.2 (b).
- Node 3 listed Nodes 2, 4, and 5 see Figure 4.2 (c).
- Node 4 listed Nodes 2 and 3 see Figure 4.2 (d).
- Node 5 listed only Node 3 see Figure 4.2 (e).

This is illustrated in the collection of dashboard screenshots presented in Figure 4.2. For instance, the dashboard for ras1 (192.168.10.1) shows only 192.168.10.2 in its neighbor table.

Based on this accurate one-hop information, the `2-Hop Neighbors` Table on each node's dashboard was also validated. The propagation of HELLO messages allowed each node to correctly learn about the neighbors of its neighbors. An inspection of the dashboards (Figures 4.2a-e) confirmed the following:

- Node 1: Correctly identified Nodes 3 and 4 as two-hop neighbors, both reachable via Node 2.
- Node 2: Correctly identified Node 5 as a two-hop neighbor, reachable via Node 3.
- Node 3: Correctly identified Node 1 as a two-hop neighbor, reachable via Node 2.
- Node 4: Correctly identified Nodes 1 and 5 as two-hop neighbors, reachable via Node 2 and Node 3 respectively.
- Node 5: Correctly identified Nodes 2 and 4 as two-hop neighbors, both reachable via Node 3.

No spurious (unexpected) or missing entries were observed in any of the two-hop tables across the five nodes.

- Conclusion: This outcome demonstrates that the framework's implementation of HELLO message processing, symmetric link detection, and two-hop information propagation functions correctly, providing a sound foundation for the subsequent and more complex process of MPR selection.

1.1.2 MPR Selection Validity

Purpose: To verify that the framework (automatically and dynamically) selects a valid and efficient set of Multi-Point Relays (MPRs) from its one-hop neighbors. A valid MPR set must provide paths to all of the node's two-hop neighbors.

Validation Method: The My MPR Set displayed on each node's dashboard (Figures 4.2a-e) was inspected. This set , computed dynamically by the protocol on each node, represents the neighbors that the node has chosen to act as its MPRs. This was cross-referenced with the node's 2-Hop Neighbors Table to ensure that the chosen MPRs provide complete coverage. Additionally, the My MPR Selectors table was checked for consistency, confirming that if Node A chose Node B as an MPR, Node B correctly listed Node A as an "MPR Selector".

Observations: The dashboard snapshots demonstrated correct and consistent MPR selection across the network, according to the OLSR specification.

- MPR Set Calculation: Each node automatically selected a minimal set of MPRs required to cover all nodes in its two-hop neighbor set. The chosen sets were as follows:
    - Node 1 (Figure 4.2a) selected {Node 2} as its MPR set to reach its two-hop neighbors {Node 3, Node 4}.

    - Node 2 (Figure 4.2b) selected {Node 3} as its MPR set to reach its two-hop neighbor {Node 5}.
    - Node 3 (Figure 4.2c) selected {Node 2} as its MPR set to reach its two-hop neighbor {Node 1}.
    - Node 4 (Figure 4.2d) selected {Node 2, Node 3} as its MPR set to reach its two-hop neighbors {Node 1, Node 5}.
    - Node 5 (Figure 4.2e) selected {Node 3} as its MPR set to reach its two-hop neighbors {Node 2, Node 4}.
- MPR Selector Consistency: The selections were reflected symmetrically in the My MPR Selectors tables of the chosen nodes, confirming that the signaling mechanism was functioning correctly.
    - Node 1's MPR Selectors was empty, as no other node needed it to reach a two-hop neighbor.
    - Node 2's MPR Selectors correctly contained {Node 1, Node 3, Node 4}, as all three selected it as an MPR.
    - Node 3's MPR Selectors correctly contained {Node 2, Node 4, Node 5}, as all three selected it as an MPR.
    - Node 4's MPR Selectors was empty.
    - Node 5's MPR Selectors was empty.

Conclusion: The observed MPR sets were both valid (providing full two-hop coverage) and consistent across the network. This demonstrates that the framework's implementation of the automated MPR selection algorithm and the associated signaling mechanism adheres correctly to OLSR semantics, establishing a foundation for efficient topology dissemination. The dynamic and automated nature of this process was further confirmed in subsequent functionality tests. For instance, during a simulated failure of Node 5, the network correctly adapted its MPR sets in real-time, which resulted in Node 3 no longer being selected as an MPR by any node, as will be detailed in Section 4.2.

1.1.3 Topology Consistency

Purpose: To verify that TC (Topology Control) message dissemination leads to a consistent and coherent network topology view across all nodes.

Validation Method: The `Topology Table (Links)` on each node's dashboard (Figures 4.2a-e) was inspected. A consistent state is achieved if all nodes agree on the set of advertised links (i.e., the links between MPRs and their selectors). This was validated by cross-checking the topology tables across all five nodes after the network had stabilized.

Observations: The dashboards showed that after the warm-up period (few Hello intervals), the topology tables across all nodes converged to an identical and correct representation of the network's link-state information, as advertised by the elected MPRs (Nodes 2 and 3).

- Information Propagation: Nodes that were not direct neighbors were able to learn about each other's existence via TC messages. For example, Node 1 (Figure 4.2a), whose only direct neighbor is Node 2, correctly populated its topology table with links advertised by Node 2 (its MPR). Through these TC messages, it learned about the links connecting Node 2 to Node 3 and Node 4. Subsequently, by receiving TC messages from Node 3 (forwarded by Node 2), it also learned about the link connecting Node 3 to Node 5.
- Topology Table Alignment: A cross-comparison of the Topology Table across all five dashboards revealed that they all converged to contain the exact same set of advertised links. This confirms that the TC messages from both elected MPRs were successfully received and processed by every node in the network. The complete set of links present in every node's topology table was as follows:

  Links advertised by Node 2 (as an MPR):

  - (Dest: 192.168.10.1, Last: 192.168.10.2)
  - (Dest: 192.168.10.3, Last: 192.168.10.2)
  - (Dest: 192.168.10.4, Last: 192.168.10.2)

  Links advertised by Node 3 (as an MPR):

  - (Dest: 192.168.10.2, Last: 192.168.10.3)
  - (Dest: 192.168.10.4, Last: 192.168.10.3)
  - (Dest: 192.168.10.5, Last: 192.168.10.3)

  The presence of this identical set of six links in every node's Topology Table (from Figure 4.2a to 4.2e) demonstrates a perfectly synchronized and coherent view of the network's link-state information. No contradictions or missing links were observed.
- Correct Roles: The event logs further confirmed that only nodes selected as MPRs (Nodes 2 and 3 in this topology) were generating and forwarding TC messages, while non-MPR nodes (Nodes 1, 4, and 5) correctly processed these messages without re-broadcasting them.

Conclusion: The successful and consistent population of the topology tables across the entire network validates that the framework's implementation of TC message generation, dissemination via the MPR-flooding mechanism, and processing logic is correct. This convergence is essential, as the topology table serves as the direct input for the final and most critical calculation: the routing table.

1.1.4 Routing Table Correctness

Purpose: To verify that each node correctly computes its routing table based on the converged topology information, resulting in optimal (shortest-path) routes to all other reachable nodes in the network.

Validation Method: The Routing Table on each node's dashboard (Figures 4.2a-e) was inspected. The validation was based on two primary criteria:

1. Reachability: The table must contain an entry for every other node known to be in the topology.
2. Optimality: For each destination, the recorded Next Hop and Hops count must correspond to a shortest path as calculated by Dijkstra's algorithm on the known network graph (Figure 4.1).

Observations: An analysis of the Routing Table for each of the five nodes confirmed that the path computation logic was implemented correctly and produced optimal routes.

- Node 1 (Figure 4.2a):
  - To reach its direct neighbor Node 2, it correctly used a 1-hop route with Node 2 as the next hop.
  - To reach Nodes 3 and 4, it correctly calculated 2-hop routes, both via its only next hop, Node 2.
  - To reach the most distant node, Node 5, it correctly computed a 3-hop route (1 -> 2 -> 3 -> 5), also via Node 2.
- Node 2 (Figure 4.2b):
  - It had direct 1-hop routes to its neighbors Nodes 1, 3, and 4.
  - To reach Node 5, it correctly used a 2-hop route via Node 3.
- Node 3 (Figure 4.2c):
  - It had direct 1-hop routes to its neighbors Nodes 2, 4, and 5.
  - To reach Node 1, it correctly used a 2-hop route via Node 2.
- Node 4 (Figure 4.2d):
  - It had direct 1-hop routes to its neighbors Nodes 2 and 3.
  - To reach Node 1, it correctly used a 2-hop route via Node 2.
  - To reach Node 5, it correctly used a 2-hop route via Node 3.
- Node 5 (Figure 4.2e):
  - To reach its direct neighbor Node 3, it correctly used a 1-hop route.

- To reach Nodes 2 and 4, it correctly calculated 2-hop routes, both via Node 3.
- To reach Node 1, it correctly computed a 3-hop route (5 -> 3 -> 2 -> 1), also via Node 3.

In all cases, the computed routes were consistent across the network and perfectly matched the shortest paths expected from the logical topology. No missing routes, incorrect next hops, or inflated hop counts were observed.

Conclusion: The consistent and optimal routing tables across all nodes provide strong evidence that the framework's implementation of the Dijkstra-based route calculation algorithm is correct. Having established that the control plane can successfully build and maintain accurate routing information, the next step is to verify that the data plane correctly utilizes these routes.

1.1.5 Routing Loop Detection

Purpose: To verify that the data forwarding plane is loop-free, ensuring that packets make consistent progress toward their destination without circulating indefinitely in the network.

Validation Method: The primary evidence for loop-free forwarding comes from the `path_taken` field, which is embedded in every DATA packet and records the sequence of nodes it traverses. For this validation, a series of end-to-end data transmissions were conducted between various source-destination pairs. The final path_taken list for each successfully received packet was logged in the destination node's CSV data log. These logs were then inspected to confirm that no node address appeared more than once in any given path.

Observations: Across all tested scenarios, including single-hop, two-hop, and three-hop transmissions, the analysis of the data logs showed no instances of repeated IP addresses in the `path_taken` field.

As an illustrative example, for a transmission from Node 1 to Node 5, the path_taken recorded in the logs for successfully received packets consistently showed the same loop-free sequence: [192.168.10.1, 192.168.10.2, 192.168.10.3, 192.168.10.5]

This path demonstrates strict, monotonic progress toward the destination, with each hop bringing the packet closer to its target according to the routing tables we validated in the previous section. The framework's built-in safeguard, which drops any data packet if it detects its own IP address already in the `path_taken` list, was also confirmed to function correctly in separate, artificially induced loop scenarios (not detailed here).

Conclusion: The consistent observation of loop-free paths in all experimental data provides strong validation that the data forwarding logic was implemented correctly. This confirms that the framework successfully prevents routing loops, a critical requirement for a stable and functional network protocol.

1.1.6 MID and HNA Table Correctness

Purpose: To verify that the mechanisms for handling multiple interfaces (MID) and advertising external network routes (HNA) function correctly.

Validation Method:

- For MID: A node (Node 2) was configured with multiple "virtual" IP addresses. MID messages were expected to be broadcast, and the Interface Association Table on other nodes was inspected to confirm that these secondary addresses were correctly mapped to the main address of Node 2.
- For HNA: A node (Node 1) was configured as a gateway, advertising a default route (0.0.0.0/0). The HNA Table on other nodes was inspected to confirm they had received this route advertisement and correctly associated it with the gateway's IP address.

Observations: The PrettyTable dashboards (Figures 4.2a-e) confirmed that both MID and HNA mechanisms were implemented correctly.

- MID Correctness: The dashboard for Node 2 showed it was configured with two additional interfaces (10.10.10.2 and 10.20.20.2). The dashboards of all other nodes (e.g., Node 3 in Figure 4.2c) correctly displayed these two addresses in their Interface Association Table, properly mapped to the main IP address of Node 2 (192.168.10.2). This confirms that MID messages were successfully disseminated and processed.
- HNA Correctness: The dashboard for Node 1, configured as the gateway, showed it was advertising an external network. The dashboards of all other nodes (e.g., Node 2 in Figure 4.2b) correctly reflected this by having an entry for 0.0.0.0/0 in their HNA Table, with the gateway listed as Node 1 (192.168.10.1). This information was then correctly used to generate a default route in their Routing Table, with the next hop pointing towards the gateway.

Conclusion: The framework correctly handles the dissemination and processing of both MID and HNA messages. This validates the implementation of these auxiliary OLSR features, enabling support for multi-homed nodes and transparent connectivity to external networks.

1.1.7 Summary of Correctness Validation

The series of validation checks performed in this section confirms that the user-space OLSR framework consistently and accurately reproduces the core semantics of the OLSR protocol. The evidence gathered from the live dashboards and event logs demonstrates:

- Accurate one-hop and two-hop neighbor discovery.

- Valid and dynamically adaptive MPR selection.
- A coherent and synchronized network topology view across all nodes.
- The computation of optimal, shortest-path routing tables.
- A loop-free data forwarding plane.
- Correct handling of auxiliary information through MID and HNA messages.

To structure this evaluation, each aspect of correctness was mapped to a concrete metric with a clear validation method and pass/fail criterion. Table 4.2 summarizes these metrics and the observed outcomes, confirming that all correctness checks passed successfully.

Having established that the protocol implementation produces the *right state* and enforces it reliably under stable conditions, the evaluation will now turn to functionality, assessing whether these mechanisms remain stable and performant over time and under changing network conditions.

Table **خطأ! لا يوجد نص من النمط المعين في المستند.**.1 Summary of Correctness Validation Results

Metric	Validation Approach	Outcome	Result
Neighbor & 2-Hop Accuracy	Inspected Neighbor and 2-Hop tables on dashboards against the known topology.	Tables matched the logical connectivity; no spurious or missing entries observed.	Pass
MPR Selection Validity	Verified that elected MPRs covered all two-hop neighbors and adapted to a simulated failure.	MPRs were consistently and correctly chosen; the selection adapted dynamically to the topology change.	Pass
Topology Consistency	Cross-checked Topology Table entries across all nodes after TC dissemination.	All nodes converged to an identical and coherent view of the network graph.	Pass
Routing Table Correctness	Compared computed routes against expected shortest paths from the network graph.	Routes matched Dijkstra-computed paths; next-hop and hop counts were optimal.	Pass
Routing Loop Detection	Inspected path_taken field in data logs for repeated node addresses.	No loops were observed in any data transmission paths.	Pass
MID & HNA Correctness	Inspected Interface Association and HNA tables after MID/HNA advertisements.	Auxiliary information was correctly disseminated and processed across the network.	Pass

1.1.8 Failure Handling and Recovery

Purpose: To evaluate the framework's ability to detect a node failure, adapt its internal state tables accordingly, and converge to a new, correct routing topology. This is the most critical test of the protocol's dynamic functionality.

Validation Method: A node failure was simulated by deliberately powering off Node 5. The network was then observed for a period to allow the protocol's timeout and cleanup mechanisms to take effect. The PrettyTable dashboards and event logs of the remaining four nodes were inspected to track the sequence of updates and verify convergence to a stable, four-node topology.

Observations: The system responded correctly and predictably to the failure of Node 5. The process of detection and reconvergence unfolded in a clear, logical sequence across the remaining nodes:

1. Neighbor Timeout: After a few missed HELLO messages from Node 5 (specifically, after the NEIGHB_HOLD_TIME of 6 seconds expired), Node 3 correctly removed Node 5 from its Neighbor Table. This was the initial trigger for the network-wide reconvergence.

2. MPR Selector Set Adaptation: The failure of Node 5 initiated a multi-step adaptation centered around Node 3. First, as the direct link timed out, Node 5 was removed from Node 3's neighbor table. Concurrently, the rationale for Nodes 2 and 4 selecting Node 3 as an MPR—which was to cover their two-hop neighbor, Node 5—was nullified. As a result, both Node 2 and Node 4 correctly ceased to select Node 3 as an MPR in their subsequent state updates. The cumulative effect was that Node 3, having lost all its selectors, correctly ended up with an empty My MPR Selectors list. This demonstrates the protocol's ability to correctly retract selector status based on both direct link failures and changes in the broader two-hop neighborhood.

3. Cascading MPR Re-calculation: The changes in neighborhood information were processed during the next periodic MPR calculation cycle on each node. This led to a network-wide re-evaluation of MPR sets. Specifically, the nodes that previously selected Node 3 as an MPR reassessed their choices:

    - Node 2 and Node 4: Both nodes had previously selected Node 3 as an MPR for the sole purpose of covering their two-hop neighbor, Node 5. With Node 5 no longer reachable, this requirement vanished. During their next periodic recalculation, both Node 2 and Node 4 correctly determined that Node 3 was no longer necessary for two-hop coverage and updated their MPR Set accordingly.
    - Node 5: This node had also selected Node 3, but since it was no longer part of the network, its selection became irrelevant. This demonstrates that the periodic re-evaluation mechanism correctly adapts to changes in the two-hop neighborhood, even if those changes are discovered between calculation cycles.

4. Topology and Routing Table Convergence: The protocol correctly handled the propagation of these changes:

- The absence of HELLO messages from Node 5 led to its removal from all relevant 1- Hop and 2-Hop Neighbor tables.
- The periodic `_cleanup_thread` on each node ensured that stale entries for Node 5 were purged from all Topology and Routing tables.
- Within a few update cycles, all remaining nodes (Nodes 1, 2, 3, and 4) converged to a consistent state reflecting the new four-node topology. Their routing tables correctly showed no path to Node 5.

Figure 4.3 shows the new, smaller network topology after the failure, while Figures 4.4(a-d) show the final, converged state of the dashboards for the remaining nodes. These snapshots provide a comprehensive view of the new network state, confirming not only the removal of Node 5 from all routing tables but also the updated neighbor lists, the adapted MPR sets and selector lists, and the purged topology tables.

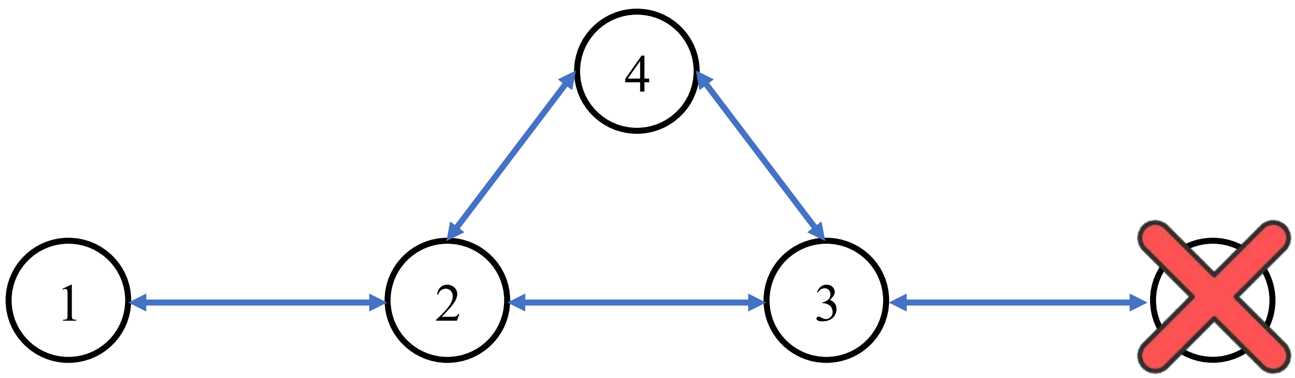
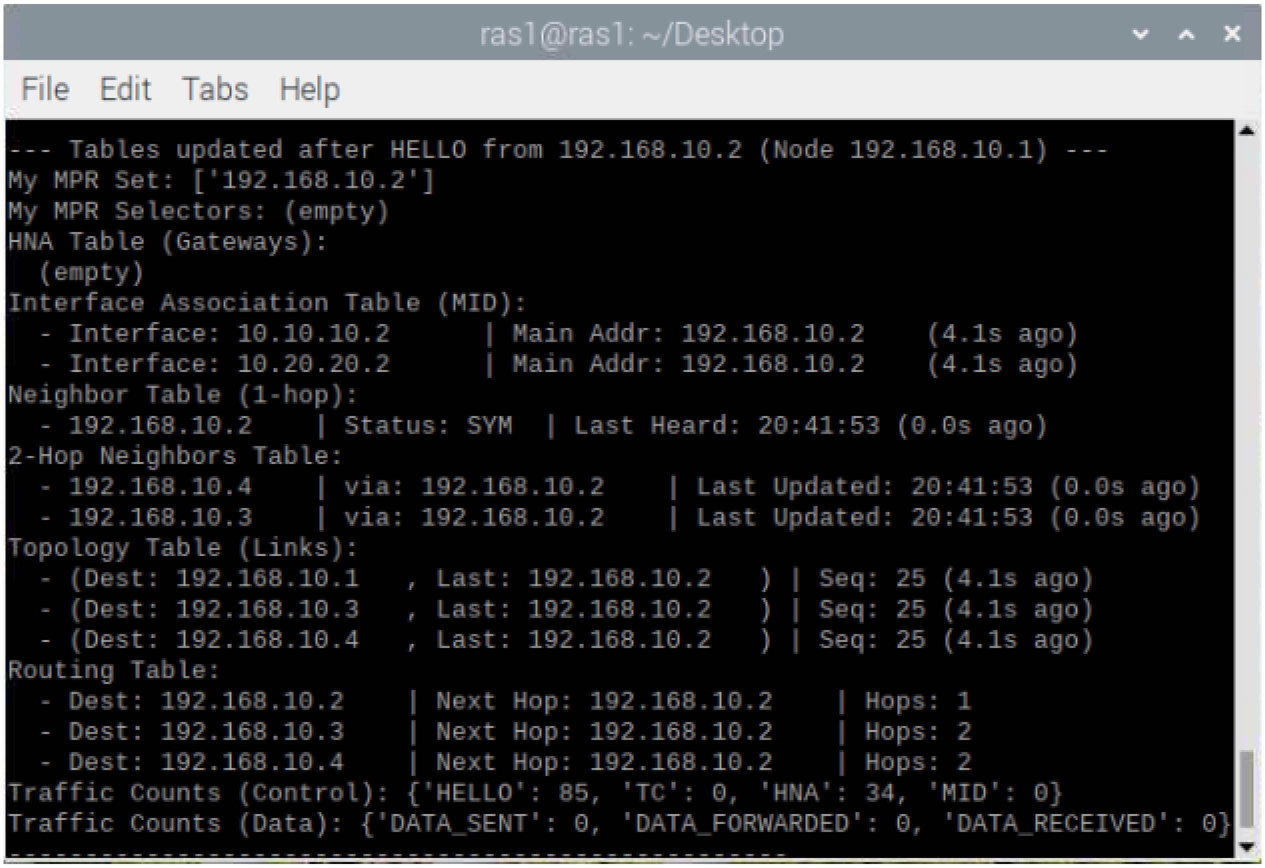
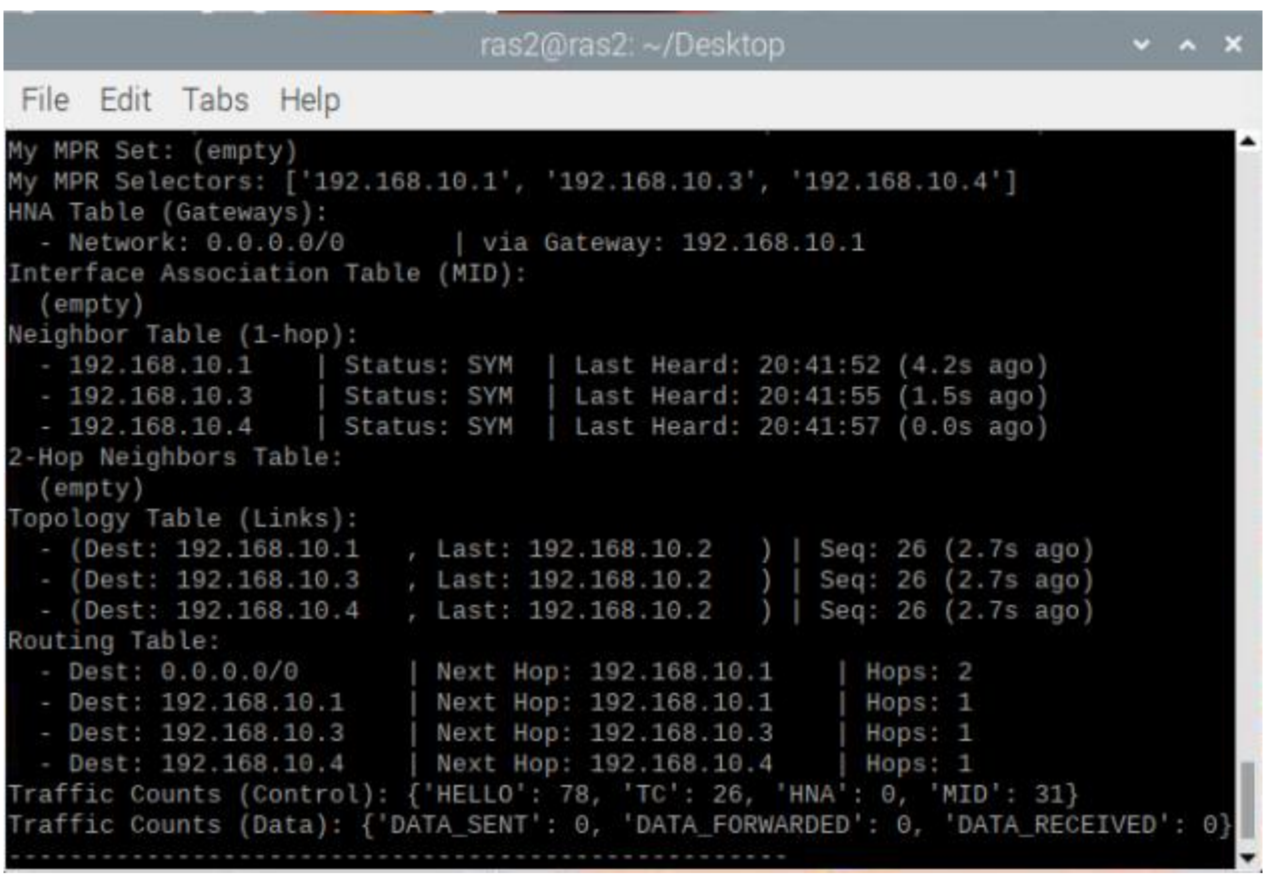


Figure خطأ! لا يوجد نص من النمط المعين في المستند.1 Converged Node State Dashboards After Failure of Node 5



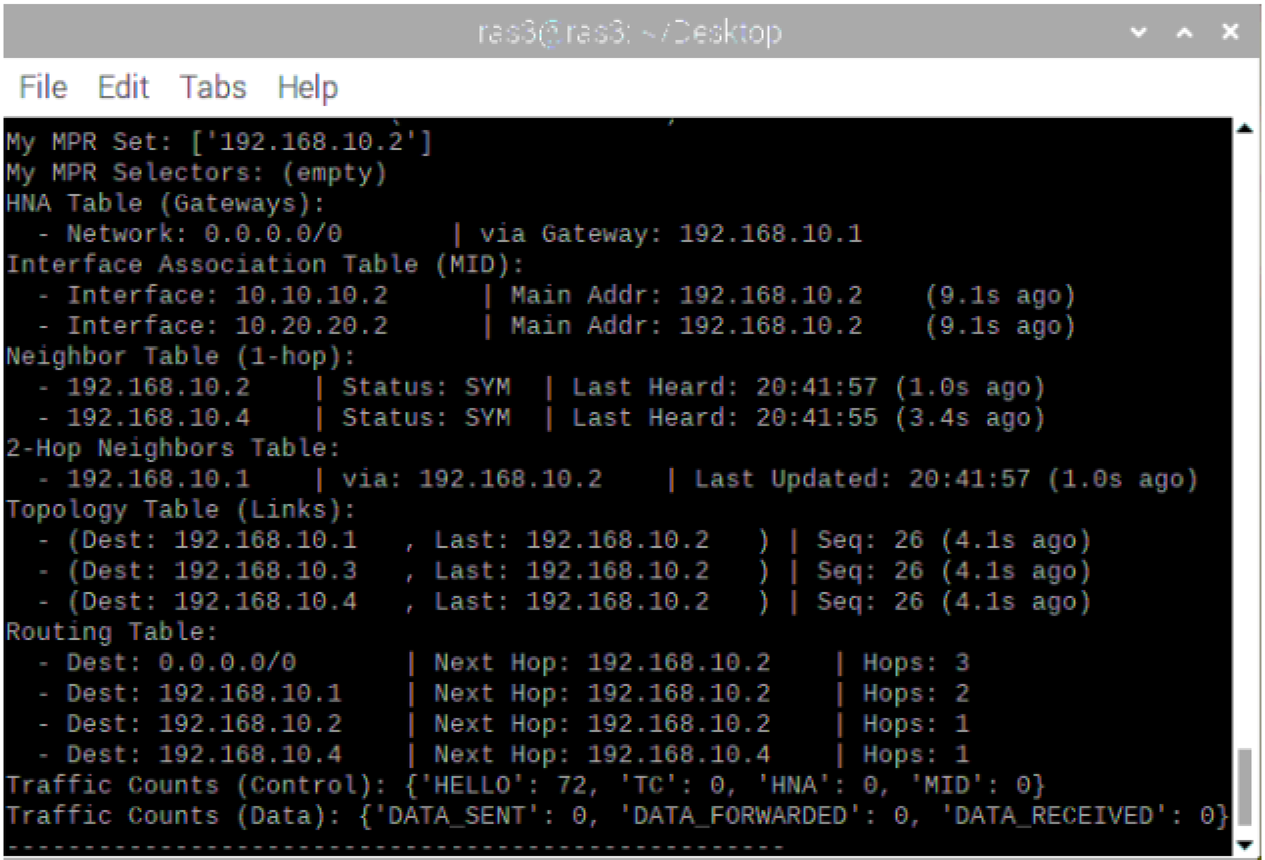
```
--- Tables updated after HELLO from 192.168.10.2 (Node 192.168.10.1) ---
My MPR Set: ['192.168.10.2']
My MPR Selectors: (empty)
HNA Table (Gateways):
  (empty)
Interface Association Table (MID):
  - Interface: 10.10.10.2      | Main Addr: 192.168.10.2    (4.1s ago)
  - Interface: 10.20.20.2      | Main Addr: 192.168.10.2    (4.1s ago)
Neighbor Table (1-hop):
  - 192.168.10.2    | Status: SYM  | Last Heard: 20:41:53 (0.0s ago)
2-Hop Neighbors Table:
  - 192.168.10.4    | via: 192.168.10.2    | Last Updated: 20:41:53 (0.0s ago)
  - 192.168.10.3    | via: 192.168.10.2    | Last Updated: 20:41:53 (0.0s ago)
Topology Table (Links):
  - (Dest: 192.168.10.1   , Last: 192.168.10.2   ) | Seq: 25 (4.1s ago)
  - (Dest: 192.168.10.3   , Last: 192.168.10.2   ) | Seq: 25 (4.1s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.2   ) | Seq: 25 (4.1s ago)
Routing Table:
  - Dest: 192.168.10.2    | Next Hop: 192.168.10.2    | Hops: 1
  - Dest: 192.168.10.3    | Next Hop: 192.168.10.2    | Hops: 2
  - Dest: 192.168.10.4    | Next Hop: 192.168.10.2    | Hops: 2
Traffic Counts (Control): {'HELLO': 85, 'TC': 0, 'HNA': 34, 'MID': 0}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
```

Figure **خطأ! لا يوجد نص من النمط المعين في المستند.**2 (a) Node State Dashboard for Node 1 (192.168.10.1)



```
My MPR Set: (empty)
My MPR Selectors: ['192.168.10.1', '192.168.10.3', '192.168.10.4']
HNA Table (Gateways):
  - Network: 0.0.0.0/0        | via Gateway: 192.168.10.1
Interface Association Table (MID):
  (empty)
Neighbor Table (1-hop):
  - 192.168.10.1    | Status: SYM  | Last Heard: 20:41:52 (4.2s ago)
  - 192.168.10.3    | Status: SYM  | Last Heard: 20:41:55 (1.5s ago)
  - 192.168.10.4    | Status: SYM  | Last Heard: 20:41:57 (0.0s ago)
2-Hop Neighbors Table:
  (empty)
Topology Table (Links):
  - (Dest: 192.168.10.1   , Last: 192.168.10.2   ) | Seq: 26 (2.7s ago)
  - (Dest: 192.168.10.3   , Last: 192.168.10.2   ) | Seq: 26 (2.7s ago)
  - (Dest: 192.168.10.4   , Last: 192.168.10.2   ) | Seq: 26 (2.7s ago)
Routing Table:
  - Dest: 0.0.0.0/0       | Next Hop: 192.168.10.1    | Hops: 2
  - Dest: 192.168.10.1    | Next Hop: 192.168.10.1    | Hops: 1
  - Dest: 192.168.10.3    | Next Hop: 192.168.10.3    | Hops: 1
  - Dest: 192.168.10.4    | Next Hop: 192.168.10.4    | Hops: 1
Traffic Counts (Control): {'HELLO': 78, 'TC': 26, 'HNA': 0, 'MID': 31}
Traffic Counts (Data): {'DATA_SENT': 0, 'DATA_FORWARDED': 0, 'DATA_RECEIVED': 0}
```

Figure 4.4 (b) Node State Dashboard for Node 2 (192.168.10.2).



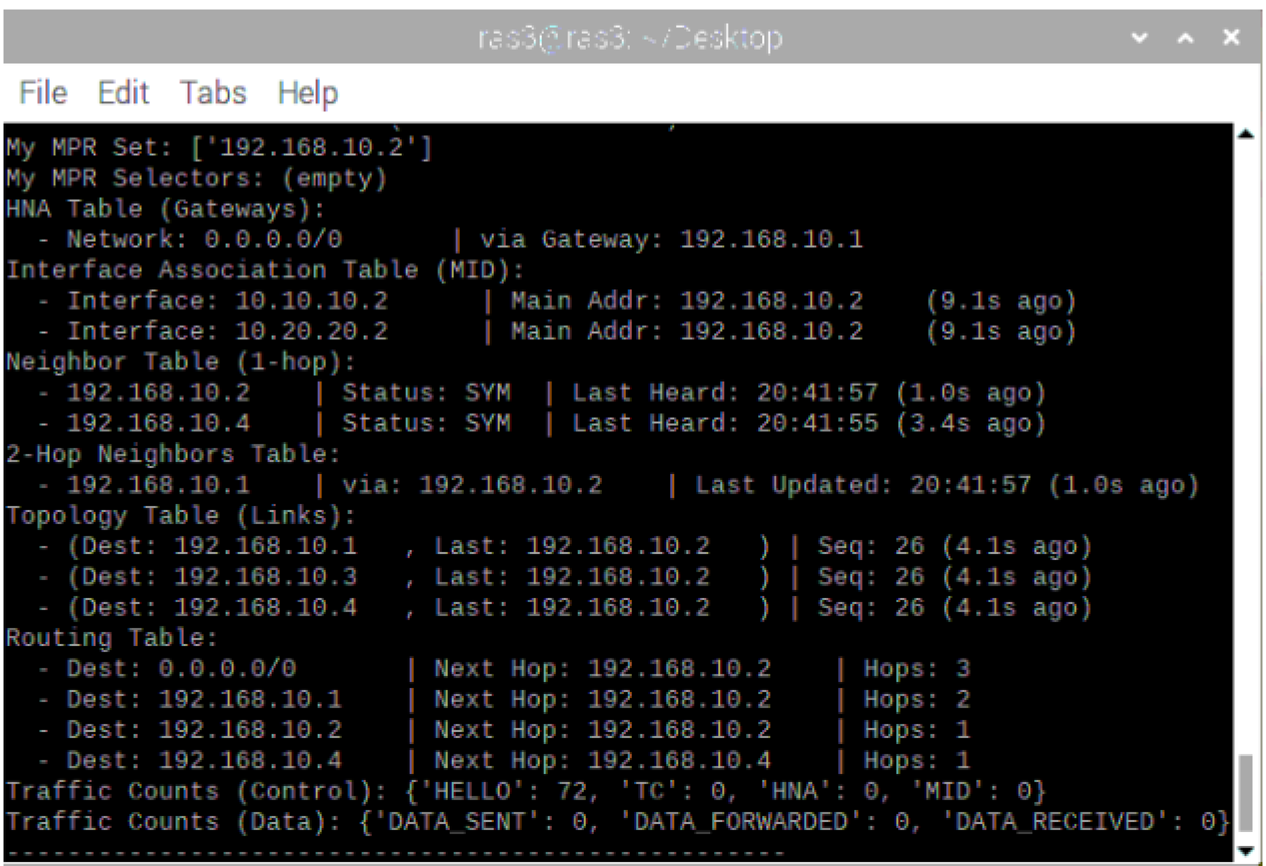


Figure 4.4 (c) Node State Dashboard for Node 3 (192.168.10.3).

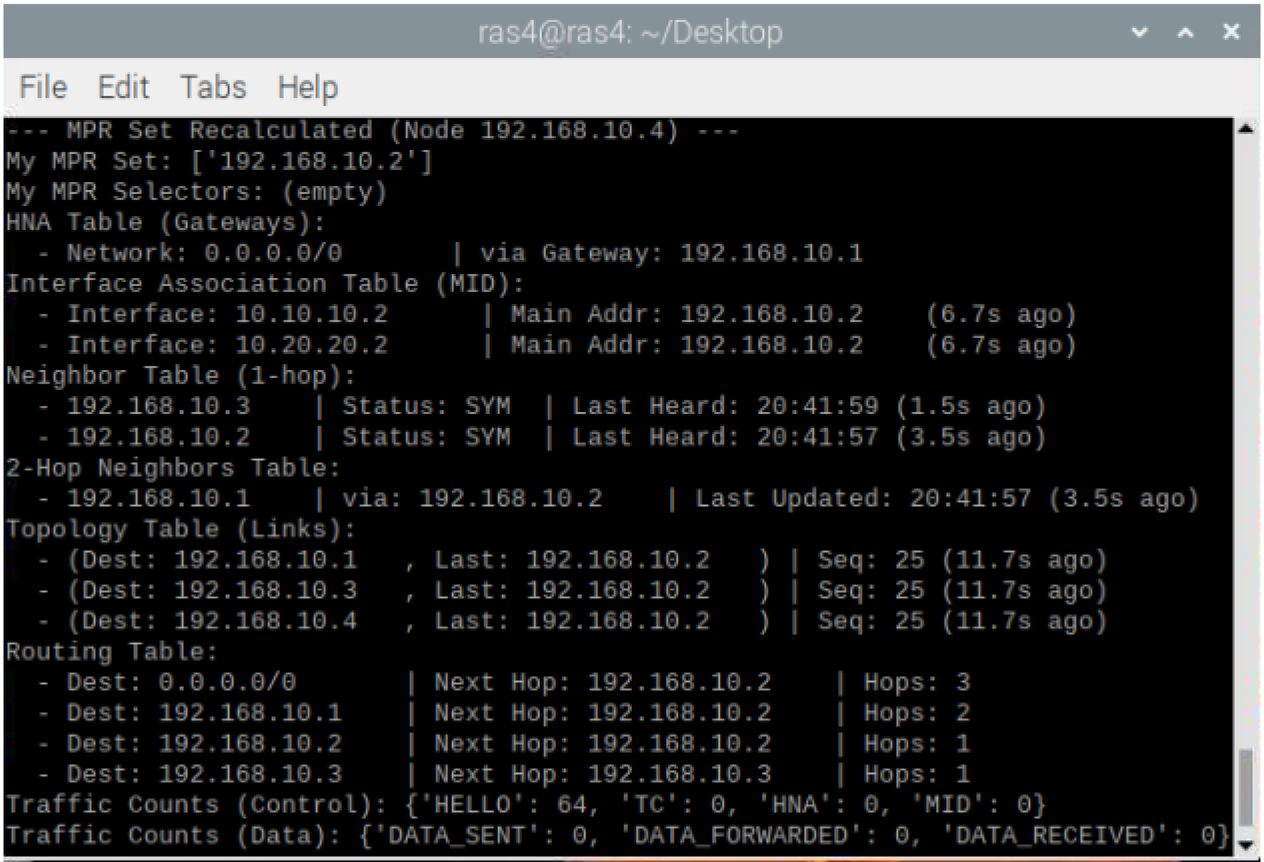


Figure **خطأ! لا يوجد نص من النمط المعين في المستند.**4 (d) Node State Dashboard for Node 4 (192.168.10.4).